Red Hat OpenShift AI Self-Managed 3.4

Managing model registries

Managing model registries in Red Hat OpenShift AI Self-Managed

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Legal Notice

Abstract

As an OpenShift AI administrator, you can create, delete, and manage permissions for model registries in OpenShift AI.

Table of Contents

PREFACE

As an OpenShift AI administrator, you can create, delete, and manage permissions for model registries in OpenShift AI.

CHAPTER 1. OVERVIEW OF THE MODEL CATALOG AND MODEL REGISTRIES

The model catalog provides a curated library where data scientists and AI engineers can discover and evaluate the available generative AI (gen AI) models to find the best fit for their use cases.

A model registry acts as a central repository for administrators, data scientists, and AI engineers to register, version, and manage the lifecycle of AI models before configuring them for deployment. A model registry is a key component for AI model governance.

1.1. MODEL CATALOG

Data scientists and AI engineers can use the model catalog to discover and evaluate the gen AI models that are available and ready for their organization to register, deploy, and customize.

The model catalog provides models from different providers that you can search, discover, and evaluate before you register models in a model registry and deploy them to a model serving runtime. Third-party gen AI models are benchmarked by Red Hat for performance and quality by using open-source evaluation datasets. You can compare performance metrics for specific hardware configurations and determine the most suitable option for deployment.

OpenShift AI provides a default model catalog that includes models from different source providers. OpenShift AI administrators can manage and govern model catalog sources to control which models are available to data scientists and AI engineers. For more information, see Manage and govern model catalog sources.

For more information about how data scientists and AI engineers discover and evaluate models in the catalog, see Working with the model catalog .

1.2. MODEL REGISTRY

A model registry is an important component in the lifecycle of an artificial intelligence/machine learning (AI/ML) model, and is a vital part of any machine learning operations (MLOps) platform or workflow. A model registry acts as a central repository, storing metadata related to machine learning models from development to deployment. This metadata ranges from high-level information like the deployment environment and project, to specific details like training hyperparameters, performance metrics, and deployment events.

A model registry acts as a bridge between model experimentation and serving, offering a secure, collaborative metadata store interface for stakeholders in the ML lifecycle. Model registries provide a structured and organized way to store, share, version, deploy, and track models.

OpenShift AI administrators can create model registries in OpenShift AI and grant model registry access to data scientists and AI engineers. For more information, see Managing model registries.

Data scientists and AI engineers with access to a model registry can use it to store, share, version, deploy, and track models. For more information, see Working with model registries.

CHAPTER 2. ENABLING THE MODEL REGISTRY COMPONENT

Before data scientists and AI engineers in your organization can work with the model registry and model catalog, you must ensure that the **modelregistry** component is enabled in OpenShift AI.

> **NOTE**
>
> The **modelregistry** component is enabled by default in a new OpenShift AI 3.4 installation. However, if the model registry was not enabled in a previous version of OpenShift AI, you must enable this component after upgrading to OpenShift AI 3.4.

Prerequisites

- You have cluster administrator privileges for your OpenShift cluster.
- You have access to the data science cluster.
- You have installed the Red Hat OpenShift AI Operator on your OpenShift cluster.
- You have sufficient resources. For more information about the minimum resources required to use OpenShift AI, see Installing and deploying OpenShift AI (for disconnected environments, see Deploying OpenShift AI in a disconnected environment).

Procedure

1. In the OpenShift console, click **Ecosystem → Installed Operators**.
2. Search for the **Red Hat OpenShift AI Operator** version 2.14+, and then click the Operator name to open the Operator details page.
3. Click the **Data Science Cluster** tab.
4. Click the default instance name (for example, **default-dsc**) to open the instance details page.
5. Click the **YAML** tab to show the instance specifications.
6. Find the **spec.components** section, and then add or update it to include the following **modelregistry** component entry, with the **managementState** field set to **Managed**, and the **registriesNamespace** field set to **rhoai-model-registries**:

```
modelregistry:
  managementState: Managed
  registriesNamespace: rhoai-model-registries
```

7. Click **Save**.

Verification

- Confirm that the model registry namespace was created successfully:
  a. In the OpenShift console, click **Home → Projects**.
  b. Confirm that the **rhoai-model-registries** namespace is displayed in the **Projects** drop-down list.

- Check the status of the **model-registry-operator-controller-manager** pod:
    a. In the OpenShift console, from the **Project** list, select **redhat-ods-applications**.
    b. Click **Workloads** → **Deployments**.
    c. Search for the **model-registry-operator-controller-manager** deployment.
    d. Check the status:
        i. Click the deployment name to open the deployment details page.
        ii. Click the **Pods** tab.
        iii. View the pod status.
        When the status of the **model-registry-operator-controller-manager-*<pod-id>*** pod is **Running**, the pod is ready to use.

Next steps

OpenShift AI administrators can create, delete, and manage permissions for model registries. For more information, see Managing model registries.

CHAPTER 3. CREATING A MODEL REGISTRY

You can create a model registry to store, share, version, deploy, and track your models.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- The model registry component is enabled in your OpenShift AI deployment. For more information, see Enabling the model registry component.
- For production use cases, you have access to one of the following external databases:
  - PostgreSQL 16.x database
  - MySQL database that uses at least MySQL 5.x. However, it is best to use MySQL 9.x.

Procedure

1. From the OpenShift AI dashboard, click **Settings → Model resources and operations→ Model registry settings**.
2. Click **Create model registry**.
   The **Create model registry** dialog opens.
3. In the **Name** field, enter a name for the model registry.
4. Optional: Click **Edit resource name**, and enter a specific resource name for the model registry in the **Resource name** field. By default, the resource name matches the name of the model registry.

   IMPORTANT

   Resource names are what your resources are labeled as in OpenShift. Your resource name cannot exceed 253 characters, must consist of lowercase alphanumeric characters or -, and must start and end with an alphanumeric character. Resource names are not editable after creation.

   The resource name must not match the name of any other model registry resource in your OpenShift cluster.

5. Optional: In the **Description** field, enter a description for the model registry.
6. In the **Database** section, click one of the following options to select where your model data is stored:
   - **Default database (non-production)**: Use the PostgreSQL database that is enabled by default on your OpenShift cluster.

     NOTE

     The default database is designed for evaluation, development, and testing purposes only. Red Hat does not support this database for production use cases.

- **External database**: Connect an external MySQL or PostgreSQL database designed for production use cases.

7. If you selected **External database**, enter the details for the external database where your model data is stored:
   - **Database type**: Select **MySQL** or **PostgreSQL** from the list.
   - **Host**: Enter the database hostname, depending on where the database is running:
     a. In the **rhoai-model-registries** namespace, enter only the hostname for the database.
     b. In a different namespace from **rhoai-model-registries**, enter the database hostname details in **<hostname>.<namespace>.svc.cluster.local** format.
   - **Port**: Enter the port number for the database. The default port for MySQL is **3306**. The default port for PostgreSQL is **5432**.
   - **Username**: Enter the default user name that is connected to the database.
   - **Password**: Enter the password for the default user account.
   - **Database**: Enter the database name.
   - Optional: **Add CA certificate to secure database connection** Select this option to use a certificate with your database connection and specify one of the following options.

     IMPORTANT

     If your external database is configured to enforce Transport Layer Security (TLS), you must add a Certificate Authority (CA) certificate.

     a. Click **Use cluster-wide CA bundle** to use the **ca-bundle.crt** bundle in the **odh-trusted-ca-bundle** ConfigMap.
     b. Click **Use Red Hat OpenShift AI CA bundle** to use the **odh-ca-bundle.crt** bundle in the **odh-trusted-ca-bundle** ConfigMap.
     c. Click **Choose from existing certificates** to select an existing certificate. You can select the key of any ConfigMap or secret in the **rhoai-model-registries** namespace.
        i. From the **Resource** list, select a ConfigMap or secret.
        ii. From the **Key** list, select a key.
     d. Click **Upload new certificate** to upload a new certificate as a ConfigMap.
        i. Drag and drop the PEM file for your certificate into the **Certificate** field, or click **Upload** to select a file from your local machine’s file system.

NOTE

Uploading a certificate creates the **db-credential** ConfigMap with the **ca.crt** key.

To upload a certificate as a secret, you must create a secret in the OpenShift **rhoai-model-registries** namespace, and then select it as an existing certificate when you create your model registry.

For more information about creating secrets, see Providing sensitive data to pods by using secrets in the OpenShift Container Platform documentation.

8. Click **Create**.

NOTE

To find the resource name or type of a model registry, click the help icon beside the registry name. Resource names and types are used to find your resources in OpenShift.

Verification

- The new model registry is displayed on the **Model registry settings** page.
- You can edit the model registry by clicking the action menu (⋮) beside it, and then clicking **Edit model registry**.

Next steps

- After creating the model registry, manage its permissions to make the model registry accessible to users and projects. For more information, see Managing model registry permissions.
- You can register a model in the model registry from the **Registry** tab. For more information, see Working with model registries.

CHAPTER 4. EDITING A MODEL REGISTRY

You can edit the details of existing model registries, such as the model registry name, description, and database connection details.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- The model registry component is enabled in your OpenShift AI deployment. For more information, see Enabling the model registry component.
- Your OpenShift AI deployment contains at least one model registry.

Procedure

1. From the OpenShift AI dashboard, click **Settings → Model resources and operations→ Model registry settings**.
2. Click the action menu (⋮) beside the model registry that you want to edit, and then click **Edit model registry**.
   The **Edit model registry** dialog opens.
3. Optional: In the **Name** field, edit the name of the model registry.
4. Optional: In the **Description** field, edit the description of the model registry.
5. Optional: In the **Database** section, click one of the following options to select where your model data is stored:
   - **Default database (non-production)**: Use the PostgreSQL database that is enabled by default on your OpenShift cluster.

     NOTE

     The default database is designed for evaluation, development, and testing purposes only. This database is not supported by Red Hat for production use cases.

   - **External database**: Connect an external MySQL or PostgreSQL database designed for production use cases.
6. If you selected **External database**, edit the details for the external database where your model data is stored:
   - **Database type**: Select **MySQL** or **PostgreSQL** from the list.
   - **Host**: Enter the database hostname, depending on where the database is running:
     a. In the **rhoai-model-registries** namespace, enter only the hostname for the database.
     b. In a different namespace from **rhoai-model-registries**, enter the database hostname details in **<hostname>.<namespace>.svc.cluster.local** format.
   - **Port**: Enter the port number for the database. The default port for MySQL is **3306**. The default port for PostgreSQL is **5432**.

- **Username**: Enter the default user name that is connected to the database.
- **Password**: Enter the password for the default user account.
- **Database**: Enter the database name.
- Optional: **Add CA certificate to secure database connection** Select this option to use a certificate with your database connection and specify one of the following options.

> **IMPORTANT**
>
> If your external database is configured to enforce Transport Layer Security (TLS), you must add a Certificate Authority (CA) certificate.

a. Click **Use cluster-wide CA bundle** to use the **ca-bundle.crt** bundle in the **odh-trusted-ca-bundle** ConfigMap.

b. Click **Use Red Hat OpenShift AI CA bundle** to use the **odh-ca-bundle.crt** bundle in the **odh-trusted-ca-bundle** ConfigMap.

c. Click **Choose from existing certificates** to select an existing certificate. You can select the key of any ConfigMap or secret in the **rhoai-model-registries** namespace.

   i. From the **Resource** list, select a ConfigMap or secret.

   ii. From the **Key** list, select a key.

d. Click **Upload new certificate** to upload a new certificate as a ConfigMap.

   i. Drag and drop the PEM file for your certificate into the **Certificate** field, or click **Upload** to select a file from your local machine's file system.

> **NOTE**
>
> Uploading a certificate creates the **db-credential** ConfigMap with the **ca.crt** key.
>
> To upload a certificate as a secret, you must create a secret in the OpenShift **rhoai-model-registries** namespace, and then select it as an existing certificate when you create your model registry.
>
> For more information about creating secrets, see Providing sensitive data to pods by using secrets in the OpenShift Container Platform documentation.

7. Click **Update**.

Verification

- The model registry is displayed with updated details on the **Model registry settings** page.

CHAPTER 5. MANAGING MODEL REGISTRY PERMISSIONS

You can manage access to a model registry for individual users and user groups in your organization, and for service accounts in a project.

> **NOTE**
>
> OpenShift AI creates the **<model-registry-name>-users** group automatically for use with model registries. You can add users to this group in OpenShift, or ask the cluster administrator to do so.
>
> The model registry operator uses OpenShift Role-Based Access Control (RBAC), and creates various RBAC resources in the **rhoai-model-registries** namespace.
>
> For each model registry instance, the operator creates a **registry-users-<model registry instance name>** role and an OpenShift group called **<model registry instance name>-users**. To grant an individual user, service account, or group access to a model registry instance, your cluster administrator must create a role binding to the **registry-users-<model registry instance name>** role for the instance.
>
> The **<model registry instance name>-users** group has a role binding to the **registry-users-<model registry instance name>** role. Your cluster administrator can add users to this group to grant them access to the model registry instance without needing to create a role binding for each user.
>
> For more information about managing RBAC in OpenShift, see Using RBAC to define and apply permissions.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- An available model registry exists in your deployment.
- The users and groups that you want to provide access to already exist in OpenShift. For more information, see Managing users and groups.

Procedure

1. From the OpenShift AI dashboard, click **Settings → Model resources and operations→ AI registry settings**.
2. Click **Manage permissions** beside the model registry that you want to manage access for. The permissions page for the model registry opens.
3. Provide one or more OpenShift groups with access to the project.
   a. On the **Users** tab, in the **Groups** section, click **Add group**.
   b. From the **Select a group** drop-down list, select a group.

      > **NOTE**
      >
      > To enable access for all cluster users, add **system:authenticated** to the group list.

c. To confirm your entry, click **Confirm** (✔).

d. Optional: To add an additional group, click **Add group** and repeat the process.

4. Provide one or more users with access to the model registry.

   a. On the **Users** tab, in the **Users** section, click **Add user**.

   b. In the **Type username** field, enter the username of the user to whom you want to provide access.

   c. To confirm your entry, click **Confirm** (✔).

   d. Optional: To add an additional user, click **Add user** and repeat the process.

5. Provide all service accounts in a project with access to the model registry.

   a. On the **Projects** tab, in the **Projects** section, click **Add project**.

   b. In the **Select or enter a project** field, select or enter the name of the project to which you want to provide access.

   c. To confirm your entry, click **Confirm** (✔).

   d. Optional: To add an additional project, click **Add project** and repeat the process.

Verification

- Users, groups, and accounts that were granted access to a model registry can register, view, edit, version, deploy, delete, archive, and restore models in that registry.
- The **Users** and **Groups** sections on the **Permissions** tab show the respective users and groups that you granted access to the model registry.
- The **Projects** sections on the **Projects** tab show the projects that you granted access to the model registry.

After you provide access to a model registry, users with access can store, share, version, deploy, and track models using the model registry feature. For more information, see Working with model registries.

CHAPTER 6. DELETING A MODEL REGISTRY

You can delete model registries that you no longer require.

IMPORTANT

When you delete a model registry, databases connected to the model registry will not be removed. To remove any remaining databases, contact your cluster administrator.

Prerequisites

- You have logged in to OpenShift AI as a user with OpenShift AI administrator privileges.
- An available model registry exists in your deployment.

Procedure

1. From the OpenShift AI dashboard, click **Settings → Model resources and operations→ AI registry settings**.
2. Click the action menu (⋮) beside the model registry that you want to delete.
3. Click **Delete model registry**.
4. In the **Delete model registry?** dialog that opens, enter the name of the model registry in the text field to confirm that you intend to delete it.
5. Click **Delete model registry**.

Verification

- The model registry is no longer displayed on the **AI registry settings** page.